

Class 17: Analyzing Sequence Data in the Cloud

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The Sequence Read Archive (SRA)

Q. What shell command can you use to view the top few files of your FASTQ file?

You can use head to view the top few files.

Q. What length are these sequence reads?

38.

Q. Can you use the grep command to determine how many total reads are in this file?

Yes: `grep -c "@SRR600956" SRR600956.fastq`

Q. Does your number of reads from grep match the name of the last read in the file?
If not why not?

Yes it matches.

Working with RNA-Seq data

Q. How would you check that these files with extension '.fastq' actually look like what we expect for a FASTQ file? You could try printing the first few lines to the shell standard output:

We use the command head

Q. How could you check the number of sequences in each file?

We use the command `grep -c "@SRR2156848" SRR2156848_1.fastq`

Q. Check your answer with the bottom of the file using the tail command and also check the matching mate pair FASTQ file. Do these numbers match? If so why or why not?

Yes, they match.

Q. Download the other 3 datasets (SRR2156849, SRR2156850 and SRR2156851) we need for our analysis, first with prefetch and then process with fasterq-dump

Done.

Q. Check you have pairs of FASTQ files for all four datasets and that they have the same number of counts in each pair?

Done.

Transcript quantification via pseudoalignment

Q. Can you run kallisto to print out its citation information?

kallisto cite.

Check your results

Q. Have a look at the TSV format versions of these files to understand their structure. What do you notice about these files contents?

It's in a structured format with target_id, length, eff_length, and est_counts, tpm. Most est_counts are 0 and very few higher in the 0-2 range.

Downstream analysis

```
library(tximport)

# setup the folder and file-names to read
folders <- dir(pattern="SRR21568*")
samples <- sub("_quant", "", folders)
files <- file.path( folders, "abundance.h5" )
names(files) <- samples

txi.kallisto <- tximport(files, type = "kallisto", txOut = TRUE)
```

1 2 3 4

```
head(txi.kallisto$counts)
```

	SRR2156848	SRR2156849	SRR2156850	SRR2156851
ENST00000539570	0	0	0.00000	0
ENST00000576455	0	0	2.62037	0
ENST00000510508	0	0	0.00000	0
ENST00000474471	0	1	1.00000	0
ENST00000381700	0	0	0.00000	0
ENST00000445946	0	0	0.00000	0

```
colSums(txi.kallisto$counts)
```

SRR2156848	SRR2156849	SRR2156850	SRR2156851
2563611	2600800	2372309	2111474

```
sum(rowSums(txi.kallisto$counts)>0)
```

```
[1] 94561
```

```
to.keep <- rowSums(txi.kallisto$counts) > 0  
kset.nonzero <- txi.kallisto$counts[to.keep,]
```

```
keep2 <- apply(kset.nonzero,1,sd)>0  
x <- kset.nonzero[keep2,]
```

Principal Component Analysis

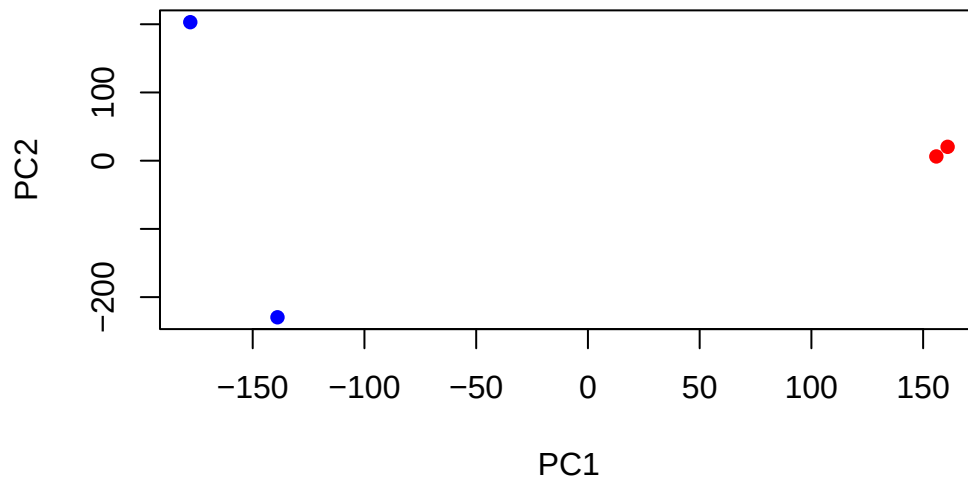
```
pca <- prcomp(t(x), scale=TRUE)
```

```
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	183.6379	177.3605	171.3020	1e+00
Proportion of Variance	0.3568	0.3328	0.3104	1e-05
Cumulative Proportion	0.3568	0.6895	1.0000	1e+00

```
plot(pca$x[,1], pca$x[,2],
     col=c("blue","blue","red","red"),
     xlab="PC1", ylab="PC2", pch=16)
```

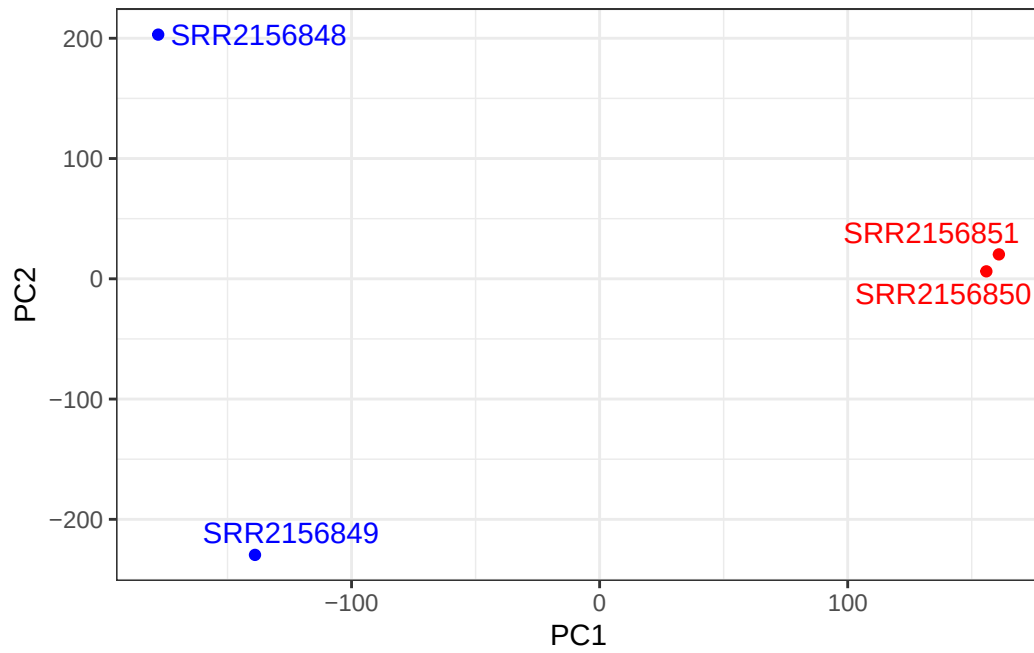


Q. Use ggplot to make a similar figure of PC1 vs PC2 and a separate figure PC1 vs PC3 and PC2 vs PC3.

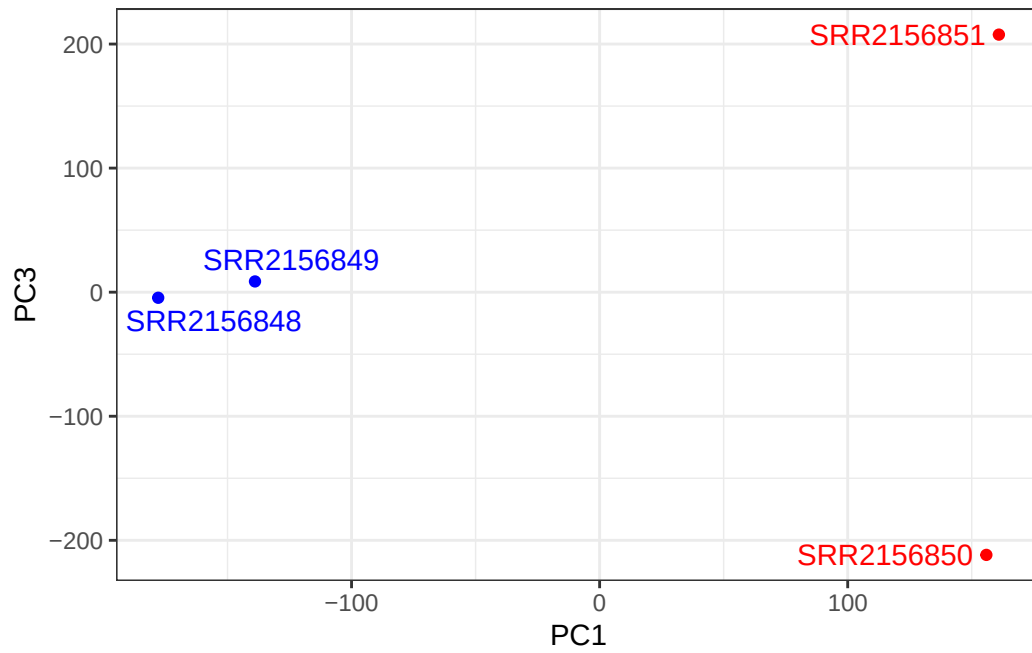
```
library(ggplot2)
library(ggrepel)

mycols <- c("blue","blue","red","red")

ggplot(pca$x) +
  aes(PC1, PC2, label=rownames(pca$x)) +
  geom_point( col=mycols ) +
  geom_text_repel( col=mycols ) +
  theme_bw()
```



```
ggplot(pca$x) +  
  aes(PC1, PC3, label=rownames(pca$x)) +  
  geom_point(col=mycols) +  
  geom_text_repel(col=mycols) +  
  theme_bw()
```



```
ggplot(pca$x) +  
  aes(PC2, PC3, label=rownames(pca$x)) +  
  geom_point(col=mycols) +  
  geom_text_repel(col=mycols) +  
  theme_bw()
```

